Clara Dehman

Department of Physics, University of Alicante, 03690 Alicante, Spain
Clara Dehman | (+34) 692 953 207 | clara.dehman@ua.es | 0000-0003-0554-7286

Education

- Nov 2023: **Ph.D. in Physics**, Autonomous University of Barcelona, Spain — Cum Laude, International Mention
- Sep 2019: **Laurea Magistrale in Fisica**, University of Padua, Italy — Cum Laude (Erasmus Mundus NucPhys Program)
- Sep 2019: **European M.Sc. in Theoretical Nuclear Physics**, Spanish Consortium Universities, Spain — Matrícula de Honor (Erasmus Mundus NucPhys Program)
- Sep 2019: **M.Sc. in Physics**, University of Caen Normandy, France (Erasmus Mundus NucPhys Program)
- Jul 2017: **Bachelor's Degree in Physics**, Lebanese University, Lebanon — Ranked 1^{st} in the Physics Department

Research Positions

- Starting Nov 2026: **API Postdoctoral Researcher**, Anton Pannekoek Institute, Netherlands — Modeling Fast Radio Bursts
- Mar 2024–Oct 2026: **Juan de la Cierva Fellow**, University of Alicante, Spain — Chiral anomalies in magnetars
- Nov 2023–Feb 2024: **Postdoctoral Researcher**, Institute of Space Sciences, Spain — 3D MHD simulations of neutron stars
- Apr–Dec 2023 **Visiting Ph.D. Fellow**, Nordita, Sweden — Inverse cascade in neutron star crusts
- Nov 2019–Nov 2023: **Doctoral Researcher**, Institute of Space Sciences, Spain — Developed the 3D code *MATINS*
- Feb–Sep 2019: **Graduate Researcher**, University of Barcelona, Spain — Master's thesis on the BCPM equation of state
- Sep 2018–Jan 2019: **Intern**, Laboratoire de Physique Corpusculaire, France — Metamodeling for neutron star equations of state
- Feb–Jun 2018: **Intern**, Legnaro National Laboratories, Italy — Measured sub-barrier fusion cross sections

Fellowships, Grants, & Funding

Total personal funding secured: €125,500 (excluding the declined €92,470 APOSTD Fellowship)

- Sept 2026–Aug 2029: **PID2025-171322NB-C21**, Agencia Estatal de Investigación, Spain — 200,000 €
- Feb 2025–Jan 2027: **Juan de la Cierva (JDC) Fellowship** (PI), Agencia Estatal de Investigación, Spain — 72,000 €
- May 2026: **COST Action SCALES Travel Grant** (PI), University of Coimbra, Portugal — 1,000 €
- Apr 2026: **International Visitor Grant** (PI), Perimeter Institute, Canada — CAD 2,000
- Oct 2024–Oct 2026: **APOSTD Fellowship** (PI), Generalitat Valenciana, Spain — 92,470 € (Declined in favor of JDC)
- Mar 2024–Dec 2026: **ASFAE/2022/26 Grant**, Generalitat Valenciana, Spain — 300,000 €
- Jan 2023–Dec 2026: **CIPROM 2022/13 Grant**, Generalitat Valenciana, Spain — 584,000 €
- Apr–Dec 2023: **Visiting Ph.D. Fellowship** (PI), Nordita, Sweden — 80,000 SEK
- Apr 2022–Dec 2024: **SGR-Cat 2021 (01269) Grant**, Generalitat de Catalunya, Spain — 60,000 €
- Jan 2020–Feb 2024: **MAGNESIA ERC Grant**, European Research Council — 2,000,000 €
- Oct 2017–Sep 2019: **EACEA Erasmus Mundus Scholarship** (PI), Joint Master Program — 37,000 €
- 2012–2014: **English Micro-Scholarship Program** (PI), U.S. Embassy in Beirut — 2,000 $

Awards & Certificates

- Feb 2026: **Seal of Excellence**, European Commission — Score: 94.6/100
- Dec 2024: **IEEC Best Doctoral Thesis Award 2024** — Awarded 1,500 €
- May 2024: **Special Mention, SEA Thesis Award 2024** — Outstanding research in a highly competitive category
- Apr 2019: **Cambridge English: Advanced (CAE)**, Cambridge Assessment English (UK) — Grade C (183/210)
- Mar 2019: **Certificate in Data Analysis and Machine Learning**, Michigan State University & University of Oslo
- Apr 2017: **ERASMUS$^+$ Joint Master Program (NucPhys)** — Selected among the top 20% of international applicants

Invited Talks & Seminars (20 total)

- Sep 2026: **Talk**, XII International Conference on Nuclear Physics in Astrophysics, Cluj-Napoca, Romania
- Jun 2026: **Review Talk**, SCALES Conference: Superfluid Condensates in Astrophysics and Laboratories, Coimbra, Portugal
- May 2026: **Seminar**, Astrophysical Fluid Dynamics Workshop, Escola d'Enginyeria de Barcelona, Spain
- Apr 2026: **Talk**, SCALES Conference: Vortex Motion from Laboratory to the Stars, Warsaw, Poland [Link]
- Apr 2026: **Strong Gravity Seminar**, Perimeter Institute, Waterloo, Canada
- Apr 2026: **Plasma-Astro Seminar**, Canadian Institute for Theoretical Astrophysics (CITA), Toronto, Canada
- Jun 2025: **Talk**, Fòrum IEEC 2025, Casa Convalescència (UAB), Barcelona, Spain
- May 2025: **Talk**, Extreme Physics of Neutron Star Interiors, Princeton University, USA
- May 2025: **Review Talk**, Workshop on Modern Equation of State and Spectroscopy in Neutron Stars, Alcalá University, Spain
- Feb 2025: **Talk**, Breaking New Ground in Supernova Physics, Fukuoka University, Japan
- Dec 2024: **Review Talk**, IReNA-INT Joint Workshop on Thermal and Magnetic Evolution, University of Washington, USA
- Sep 2024: **Talk**, Frontier Workshop in Astrophysics IV, Palermo, Italy
- Jun 2023: **Astrophysics Seminar**, University of Valencia, Spain
- May 2023: **Astrophysics Seminar**, Nordita, Sweden [Link]
- Jun 2022: **Astrocoffee Seminar**, Goethe University, Germany [Link]
- May 2022: **Seminar**, NucPhys Master Program, GANIL, France

- Dec 2021: **Seminar**, Hadronic, Nuclear & Atomic Physics Group, University of Barcelona, Spain [Link]
- Nov 2021: **Astrophysics Seminar**, University of Alicante, Spain
- Dec 2020: **Talk**, Stars & Compact Objects Meeting, Flatiron Institute, New York, USA
- Mar 2019: **Seminar**, Hadronic, Nuclear & Atomic Physics Group, University of Barcelona, Spain

Invited Scientific Visit

- Apr 2026: **Invited Scientific Visit**, Perimeter Institute for Theoretical Physics, Ontario, Canada

Contributed Talks & Seminars (13 total)

- Jun 2026: **Talk**, The X-ray Universe 2026, Elche, Spain
- Jun 2026: **Talk**, NewAthena Rising, ICE-CSIC Barcelona, Spain
- Jun 2025: **Talk**, EAS Annual Meeting, University College Cork, Ireland
- Jul 2024: **Talk**, 10^{th} International Conference on Quarks and Nuclear Physics, University of Barcelona, Spain
- Jun 2024: **Talk**, XMM-Newton Workshop 2024, European Space Agency, Madrid, Spain
- May 2024: **Special Seminar**, Institute of Space Sciences, Barcelona, Spain
- Apr 2024: **Astrophysics Seminar**, University of Alicante, Spain
- May 2022: **Talk**, Pharos Conference, University of La Sapienza, Rome, Italy
- Mar 2022: **Seminar**, Institute of Space Sciences, Barcelona, Spain [Link]
- Nov 2021: **Talk**, IAU Symposium 363 (online)
- Jun 2021: **Talk**, EAS Annual Meeting (online)
- Nov 2020: **Seminar**, Institute of Space Sciences, Barcelona, Spain [Link]
- Mar 2020: **Talk**, PHAROS Conference, Patras, Greece (canceled due to COVID-19)

Software Development & Open Access Tools

Since Sep 2020: Lead developer of `MATINS` — a 3D code for the MAgneto–Thermal evolution in Isolated Neutron Stars — `MATINS` webpage

- *Open access:* github.com/ice-csic-astroexotic/MATINS
- *Lead developers:* C. Dehman, D. Viganò, S. Ascenzi
- *Implementation:* Written in Fortran 90 with OpenMP parallelization, optimized for Intel compilers (better performance than GNU).

Mentoring

Supervision

- Since Sept 2026: **Supervisor**, B.Sc. student, University of Alicante, Spain — Ahlam Makboul
- Since Mar 2025: **Co-supervisor**, Ph.D. student, University of Alicante, Spain — Joan Llorens Ripoll
- Oct 2025–Apr 2026: **Research mentor**, M.Sc. student, Institute of Space Sciences, Spain — Riccardo Ricci
- Jan–Jul 2021: **Research mentor**, M.Sc. student, Institute of Space Sciences, Spain — Marco Ermolli

Teaching (112 Hours Total)

- 2024–2025: **Instructor**, Introduction to Modeling in Physics, University of Alicante, Spain — 45 hr (95 B.Sc. students)
- 2024–2025: **Instructor**, Nuclear and Particle Physics, University of Alicante, Spain — 54 hr (112 B.Sc. students)
- Jan–Mar 2025: **Lecturer**, Stellar Astrophysics, University of Alicante, Spain — 12 hr (14 B.Sc. students)
- May 2025: **Invited Lecturer**, Research in Computational Astrophysics, University of Alicante, Spain — 1 hr (10 M.Sc. students)

Scientific Community Service

Scientific Meeting Organization

- Sep 2026: **SOC & LOC**, 22^{nd} Pencil Code User Meeting, University of Alicante, Spain [Link]
- Nov 2022: **LOC**, Athena X-ray Observatory Conference, Barcelona, Spain [Link]
- May 2022: **LOC**, PHAROS Final Conference on Neutron Stars, University of La Sapienza, Rome, Italy [Link]

Reviewing & Evaluation Activities

- Jan 2026: **Ph.D. Committee Member**, Juan A. Gil Granados, University of Barcelona, Spain [Link]
- Since Jun 2025: **Expert Evaluator**, Ph.D. proposals, Valencian Agency for Assessment and Prospective, Spain (2 proposals)
- Since 2022: **Journal Reviewer**, Nat. Astron.; Nat. Commun.; ApJ; MNRAS; J. Phys. G

Outreach

- Jun 2026: **Social Media Outreach Coordinator**, SCALES COST Action CA24139 (WP4)
- Feb 2026: **Invited Outreach Talk**, International Day of Girls and Women in Science, University of Alicante, Spain
- Jun 2024: **Invited Interview**, on how scientific discoveries will shape the future of science, University of Elche, Spain [Link]
- Dec 2023: **Invited Interview**, on my Ph.D. experience, ICE-CSIC, Barcelona, Spain [Interview | Newsletter]
- Nov 2022: **Astronomical Observation Event**, public stargazing event, Autonomous University of Barcelona, Spain [Link]
- Sep 2022: **Research Night**, astronomical observation event for school children, Igualada, Spain
- Feb 2022: **Outreach Talk**, International Day of Girls and Women in Science, ICE-CSIC, Spain [Link]
- 2021–2023: **Magnet ICE Outreach**, collaboration with teachers at Gabriel Castellà i Raich to develop science projects promoting student integration in a school facing segregation challenges, Igualada, Spain

Press Releases (14 Total)

- Jul 2024: **Three unusually cold young neutron stars discovered**, Europa Press [Link]
- Jul 2024: **Unusually cold neutron stars detected by UA researchers**, TodoAlicante [Link]
- Jul 2024: **UA researchers contribute to discovery of unusually cold neutron stars**, University of Alicante [Link]
- Jun 2024: **Chandra peers into the densest and weirdest stars**, NASA [Link]
- Jun 2024: **Too young to be so cool: lessons from three neutron stars**, ESA [Link]
- Jun 2024: **Three neutron stars too cold for their age discovered**, IEEC [Link]
- Jun 2024: **Too young to be so cold**, MEDIA INAF [Link]
- Jun 2024: **Three neutron stars too cold for their age discovered**, ICE-CSIC [Link]
- Dec 2022: **Outburst of the young magnetar Swift J1818.0–1607 observed**, Phys.org [Link]
- Jun 2020: **Youngest neutron star discovered**, The Independent [Link]
- Jun 2020: **Youngest "baby" dead star discovered**, CNET [Link]
- Jun 2020: **Baby pulsar and rare magnetar discovered**, Astronomy Now [Link]
- Jun 2020: **Youngest magnetar ever discovered**, Sci.News [Link]
- Jun 2020: **Discovery of a "baby" star explaining cosmic explosions**, La Vanguardia [Link]

Collaborations & Memberships

- Since Jul 2026: **Junior Member**, International Astronomical Union (IAU)
- Since Nov 2025: **Member**, SCALES European COST Action CA24139
- Since Mar 2025: **Member**, NewAthena Science Community
- Since Sep 2022: **Member**, Einstein Telescope Collaboration
- Since Mar 2022: **Member**, AIHUB — Artificial Intelligence Initiative of the Spanish National Research Council (CSIC)
- 2019–2022: **Member**, Pharos European COST Action CA16214

Languages

- Arabic: Native
- English: Fluent (C2; Cambridge Advanced 183/210, TOEFL 805/900)
- French: Fluent (C2)
- Spanish: Intermediate (B2)
- Italian: Intermediate (B2)

List of Publications

Curated selection of 23 peer-reviewed publications: 8 as first author, 2 as co-first author (including a *Nature Astronomy* article), and one invited Living Review. Full publication list available via ADS, arXiv, ORCID, and Google Scholar. **Citations:** 660+ | **h-index:** 15

23. M. Ronchi, C. Pardo Araujo, V. Graber, N. Rea, C. Dehman et al. — *Connecting radio pulsars, magnetars and XDINSs in a unified evolutionary framework, using simulation-based inference.* Under minor revision in ApJ, 2026 [PDF]

22. C. Dehman — *Magnetar field dynamics driven by chiral anomalies without magnetic helicity.* In press, *Phys. Rev. D*, 2026 [arXiv:2605.08068 | DOI | cit: 2]

21. J.A. Pons, C. Dehman, D. Viganò: 2026. *Magnetic, thermal and rotational evolution of isolated neutron stars.* Living Rev Comput Astrophys 12, 5. [arXiv:2509.06699 | DOI | cit: 142]

20. A. Suvorov, C. Dehman*, J.A. Pons: 2026. *Late-blooming magnetars: awakening as ultra-long period objects after a dormant cooling epoch.* ApJ 1000, 55 (equal contribution by first two authors; CA: C. Dehman) [arXiv:2505.05373 | DOI | cit: 7]

19. A. Suvorov, C. Dehman, J.A. Pons: 2025. *Revealing the nature of ultra-long period objects with space-based gravitational-wave interferometers.* ApJ 991, 134 [arXiv:2505.06125 | DOI | cit: 6]

18. C. Dehman and J.A. Pons: 2025. *Magnetar field dynamics shaped by chiral anomalies and helicity.* Phys. Rev. Research 7, 033231 [arXiv:2505.06196 | DOI | cit: 9]

17. C. Dehman and A. Brandenburg: 2025. *Reality of inverse cascading in neutron star crusts.* A&A 694, A39 [arXiv:2408.08819 | DOI | cit: 9]

16. S. Ascenzi, D. Viganò, C. Dehman et al.: 2024. *3D code for MAgneto–Thermal evolution in Isolated Neutron Stars, MATINS: thermal evolution and light curves.* MNRAS 533, 201 [arXiv:2401.15711 | DOI | cit: 27]

15. C. Dehman, M. Centelles, X. Viñas: 2024. *Impact of the hot inner crust on compact stars at finite temperature.* A&A 687, A236 [arXiv:2401.16957| DOI | cit: 7]

14. A. Marino*, C. Dehman*, K. Kovlakas*, N. Rea* et al.: 2024. *Constraints on the dense matter equation of state from young and cold isolated neutron stars.* Nature Astron. 8, 1020 *(equal contribution by these authors)* [arXiv:2404.05371 | DOI | cit: 32]

13. J. Urbàn, P. Stefanou, C. Dehman, J.A. Pons: 2023. *Modeling force-free neutron star magnetospheres using physics-informed neural networks.* MNRAS 524, 32 [arXiv:2303.11968| DOI | cit: 20]

12. C. Dehman, D. Viganò, S. Ascenzi et al.: 2023. *3D evolution of neutron star magnetic-fields from a realistic core-collapse turbulent topology.* MNRAS 523, 5198 [arXiv:2305.06342| DOI | cit: 31]

11. C. Dehman, J.A. Pons, D. Viganò, N. Rea: 2023. *How bright can old magnetars be? Assessing the impact of magnetized envelopes and field topology on neutron star cooling.* MNRAS Lett. 520, L42 [arXiv:2301.02261| DOI| cit: 15]

10. A.Y. Ibrahim, A. Borghese, N. Rea et al.: 2023. *Deep X-ray and radio observations of the first outburst of the young magnetar Swift J1818.0-1607.* ApJ 943,20 [arXiv:2211.12391 | DOI| cit: 16]

9. C. Dehman, D. Viganò, J.A. Pons, N. Rea: 2023. *3D code for MAgneto–Thermal evolution in Isolated Neutron Stars, MATINS: the magnetic field formalism.* MNRAS 518, 1222 [arXiv:2209.12920| DOI | cit: 39]

8. N. Rea, F. Coti Zelati, C. Dehman et al.: 2022. *Constraining the nature of the 18 min periodic radio transient GLEAM-X J162759.5-523504.3 via multi-wavelength observations and magneto-thermal simulations.* ApJ 940, 72 [arXiv:2210.01903| DOI | cit: 40]

7. F. Anzuini, A. Melatos, C. Dehman et al.: 2022. *Thermal luminosity degeneracy of magnetized neutron stars with and without hyperon cores.* MNRAS 515, 3014 [arXiv:2205.14793| DOI| cit: 11]

6. F. Anzuini, A. Melatos, C. Dehman et al.: 2022. *Fast cooling and internal heating in hyperon stars.* MNRAS 509, 2609 [arXiv:2110.14039 | DOI | cit: 28]

5. D. Viganò, A. Garcia-Garcia, J.A. Pons, C. Dehman et al.: 2021. *Magneto-thermal evolution of neutron stars with coupled Ohmic, Hall and ambipolar effects via accurate finite-volume simulations.* Comput. Phys. Commun. 265, 108001 [arXiv:2104.08001 | DOI| cit: 61]

4. A.M. Stefanini, G. Montagnoli, M. D'Andrea, M. Giacomin, C. Dehman et al.: 2021. *New insights into sub-barrier fusion of* $^{28}Si + {}^{100}Mo$. J. of Phys. G 48, 055101 [ADS | DOI | cit: 21]

3. F. Coti Zelati, A. Borghese, G.L. Israel et al.: 2021. *The new magnetar SGR J1830-0645 in outburst.* ApJ Lett. 907, L34 [arXiv:2011.08653 | DOI | cit: 24]

2. C. Dehman, D. Viganò, N. Rea et al.: 2020. *On the rate of crustal failures in young magnetars.* ApJ Lett. 902, L32 [arXiv:2010.00617 | DOI | cit: 43]

1. P. Esposito, N. Rea, A. Borghese et al.: 2020. *A very young radio-loud magnetar.* ApJ Lett. 896, L30 [arXiv:2004.04083 | DOI | cit: 80]